unlikely unlicensed technologies would represent the majority of broadband connections in the near future. Even if they take a small share of the market, however, wireless last-mile systems would foster significant competition and innovation. Wired broadband providers would have to improve their offerings or lower costs to compete.

Demand Pull

Unlicensed wireless technologies will impact the broadband market even if they aren't used as the primary connection method. Large numbers of WiFi access points are being deployed for home networking. Users install these devices in their homes to share broadband connections among several computers, share peripherals such as printers, or give themselves untethered Internet access anywhere in their house.

Digital consumer electronics devices are beginning to incorporate WiFi connections as well. For example, Moxi Digital, which recently merged with Digeo, a company funded by Microsoft co-founder Paul Allen, incorporates an 802.11a transmitter in its personal media center. This allows the Moxi box to stream high-quality audio video among TVs and stereos throughout a house. Intel is spearheading a standards effort to allow WiFi to interoperate with FireWire (IEEE 1394), a wired standard popular for digital media applications.

As home networks and related devices proliferate, they will create a "pull" for broadband applications. WiFi hardware is becoming sufficiently cheap for hardware manufacturers to include it without significant effects on device prices. As users buy laptops and consumer electronics hardware with high-speed wireless connections built in, they will find new uses for it, such as sending music files from their computer to their stereo system, or sharing pictures downloaded from the Web. Many of these applications will benefit from broadband connections into the home. Wireless devices will therefore stimulate broadband demand even when the last-mile connection is wired.